

Project Proposal

VulnVerse

1. Project Title:

VulnVerse: A Comprehensive Platform for Learning and Testing Web & Network Security

2. Description:

VulnVerse is a web-based platform focused on educating users about cybersecurity through hands-on labs and integrated tools. Inspired by platforms like PortSwigger, VulnVerse extends beyond web application security by including Wi-Fi attack simulations and API testing challenges. It features custom labs for OWASP vulnerabilities, real-time tools like Nmap and hash decoders, and an AI-powered chatbot to assist users in understanding concepts or debugging code snippets. The platform aims to be an all-in-one resource for security learners.

3. Objective:

- To provide a guided, hands-on learning experience in cybersecurity.
- To simulate real-world scenarios covering web apps, Wi-Fi, and APIs.
- To integrate offensive security tools in-browser for practice.
- To offer an intelligent chatbot for query resolution and concept explanation.

4. Features and Limitations of Existing Applications:

- Name of existing application: PortSwigger Web Security Academy
- Features:
 - Labs for web vulnerabilities
 - Tutorial-based learning
 - Secure browser-based execution
- Limitations:
 - No Wi-Fi or API testing modules
 - No integration of tools like Nmap, hash decoders
 - No AI support or real-time chat assistance
 - Static lab environments, not customizable

5. Technology to be Used for Implementation:

Software:

- Frontend: HTML, Tailwind CSS, JavaScript
- Backend: Node.js with Express / Django
- Database: MongoDB / MySQL
- Chatbot: OpenAI API or LLM using HuggingFace / LangChain
- Containerization: Docker for isolated lab environments
- API Testing: Postman-like interface or Swagger simulation
- Wi-Fi Sim: Simulated attack flows using animations & mock command line

Hardware:

- Development PC (Minimum 8GB RAM recommended)
- Localhost / Cloud VM / Docker for deployment

Languages & Tools:

- JavaScript, Python, Bash
- Nmap, Hashcat, Aircrack-ng (simulated)
- AI model integration for chatbot

6. Methodology:

- Requirement Gathering: Research on existing platforms, OWASP Top 10, Wi-Fi attack vectors, and API vulnerabilities.
- Requirement Analysis: Finalize modules (Web, Wi-Fi, API, Tools, Chatbot).
- Development Phases:
 - Phase 1: UI/UX and basic platform setup
 - Phase 2: Web vulnerability labs (XSS, SQLi, IDOR, etc.)
 - Phase 3: Tool integration (Nmap, Hash Decoder, etc.)
 - Phase 4: API Testing lab (Broken Auth, Rate Limiting, etc.)
 - Phase 5: Wi-Fi attack simulation (Evil Twin, WPA2 cracking demo, etc.)
 - Phase 6: AI Chatbot for code explanation & concept query
- Testing: Manual and automated for each module
- Deployment: On local server or remote VPS
- Maintenance: Weekly updates, feedback-based iterations

7. Main Features of your Proposed App:

- ☑ Web Vulnerability Labs – Hands-on OWASP Top 10 practices
- ☑ WiFi Security Simulation – Demonstration of attacks like WPA2 handshake capture (simulated)
- ☑ API Testing Labs– Practice on REST API vulnerabilities
- ☑ Built-in Tools:
 - Nmap Scanner (via terminal UI)
 - Hash Decoder (MD5, SHA1, SHA256)
 - Request Builder (like Postman)
- ☑ AI-Powered Chatbot – Explains code, resolves doubts, and assists in labs
- ☑ Progress Tracking Dashboard– Track completed labs, scores, and learning curve
- ☑ Modular Design– Easy to add more labs or tools